

How many people can ask at once?

Live load on the shared demo: fresh questions, arrival-rate storms, and repeated FAQs. Charts show latency, success rate, and cache — not marketing targets.

RUN ID

2026-09-29-highload

PROTOCOL

07-highload-protocol

DEMO<https://demo.edgequake.com>**VERSION**

0.28.3

TENANT

00000000-0000-0000-0000-000000000002

WORKSPACE

00000000-0000-0000-0000-000000000003

Bottom line

- At 12 concurrent fresh askers, 75% finished inside ~25s (typical 22.7 s, upper 25.8 s).
- Last comfortable level: 8 concurrent users at 100% attainment.
- Repeated FAQ asks at C=32: 100% answer-cache hits, typical 9.8 s.

1. Test topology

Load was generated from a separate client machine over the public internet. Client CPU is *not* demo capacity.



Load generator (client)

Raphael-Mac-Book-Pro-Max.local

macOS 27.0 · arm64

Apple M4 Max

16 logical CPUs · 128.0 GB RAM

Python 3.12.14 · harness bench153 @ 9aa47804c

Egress IP: 194.5.82.147

System under test (demo)

<https://demo.edgequake.com>

Product 0.28.3 · storage postgresql

Postgres 18 · pgvector 0.8.5 · AGE 1.8.0

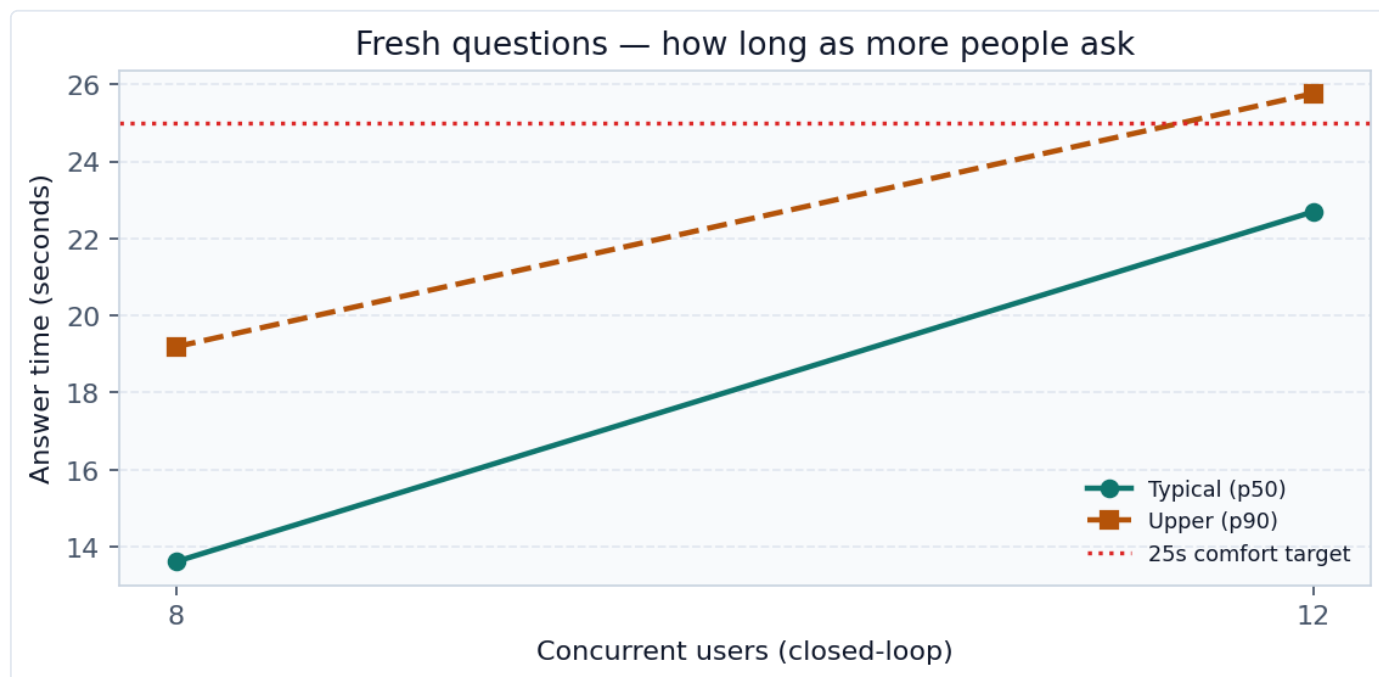
Fleet LLM: {'name': 'openai', 'model': 'gpt-5.4-mini'}

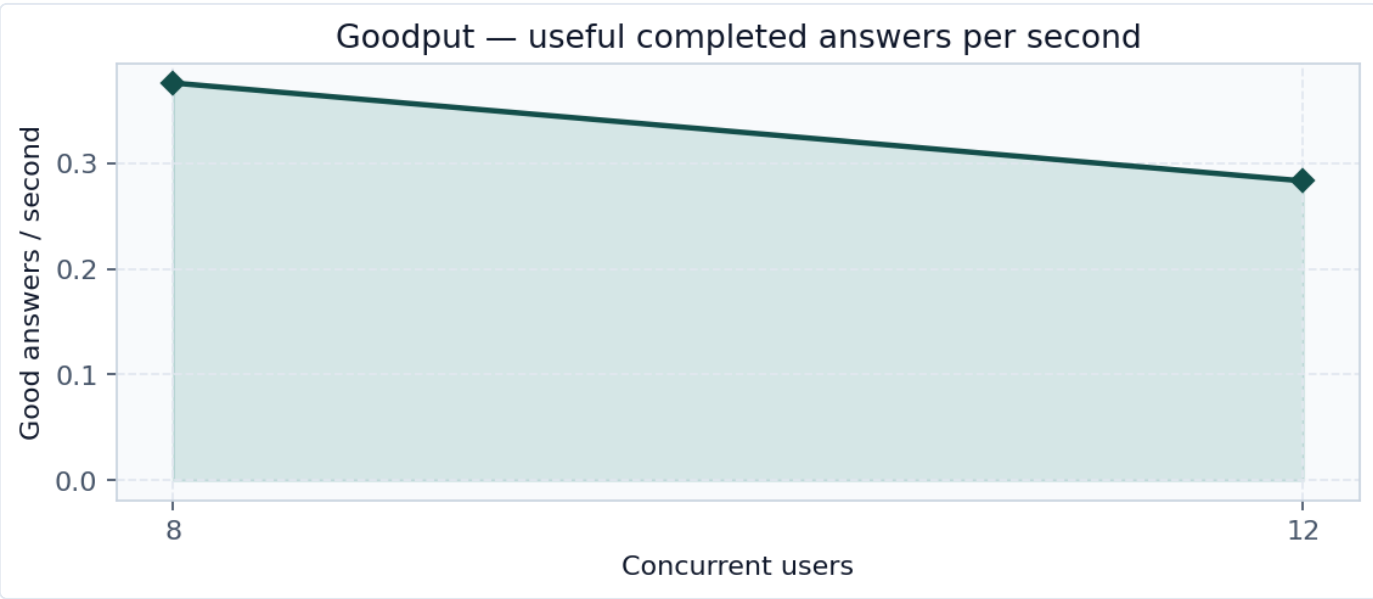
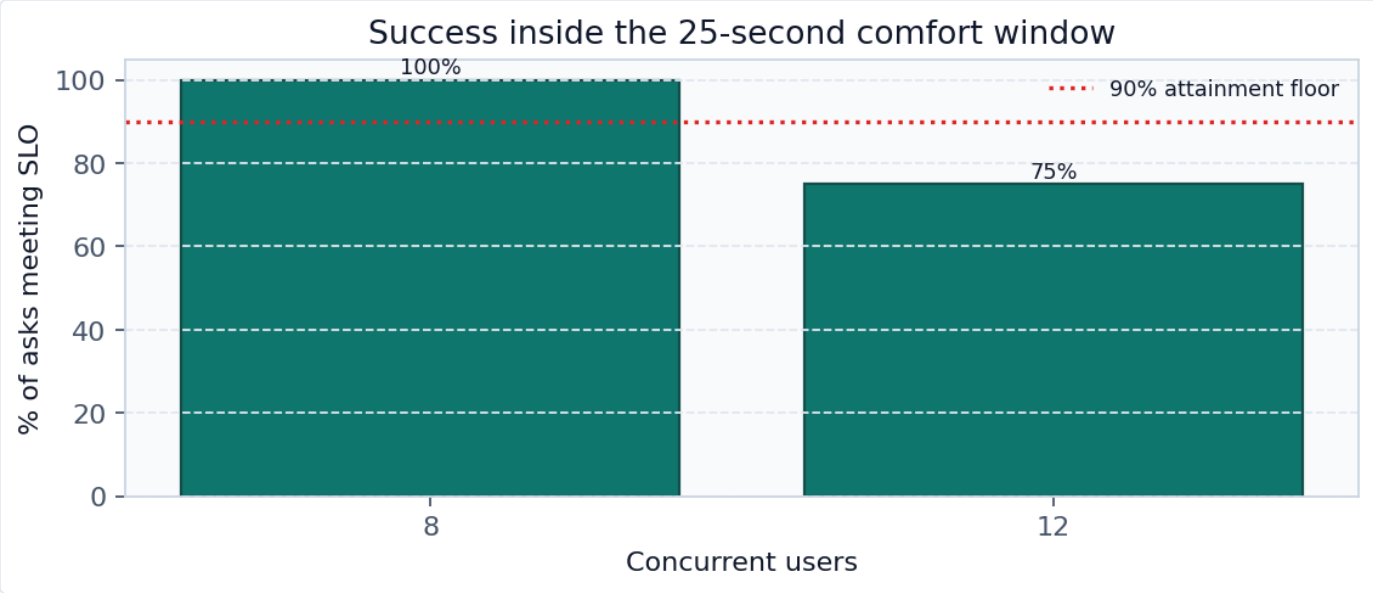
Answer LLM observed: ['mistral-small-latest']

GCP-hosted shared demo (deploy/gcp). Exact VM SKU may be opaque from outside; record only what /health and public deploy docs expose.

2. Fresh questions under concurrency (H1)

Closed-loop: a fixed number of “users” each ask a *unique* question so the answer cache should miss. Comfort target: finish within **25 seconds**. Success floor: **90%** of asks.



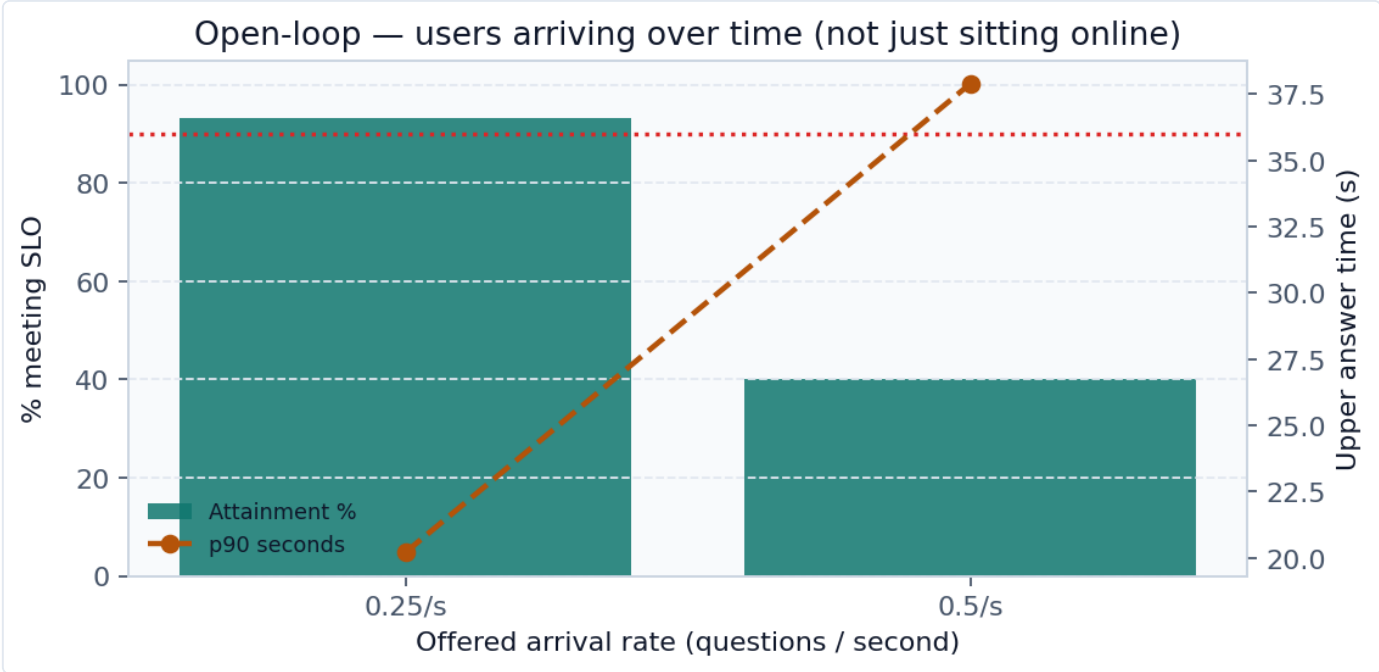


USERS	ASKS	OK	IN SLO	TYPICAL	UPPER	GOOD/S	CACHE
8	12	100%	100%	13.6 s	19.2 s	0.3758	0%
12	12	100%	75%	22.7 s	25.8 s	0.2831	0%

Knee (H1): Last comfortable level: 8 concurrent users at 100% attainment.

3. Arrival-rate storm (H2)

Open-loop: new questions arrive at a steady rate (λ asks/second), whether or not prior asks finished. This stress-tests queuing when users keep coming.

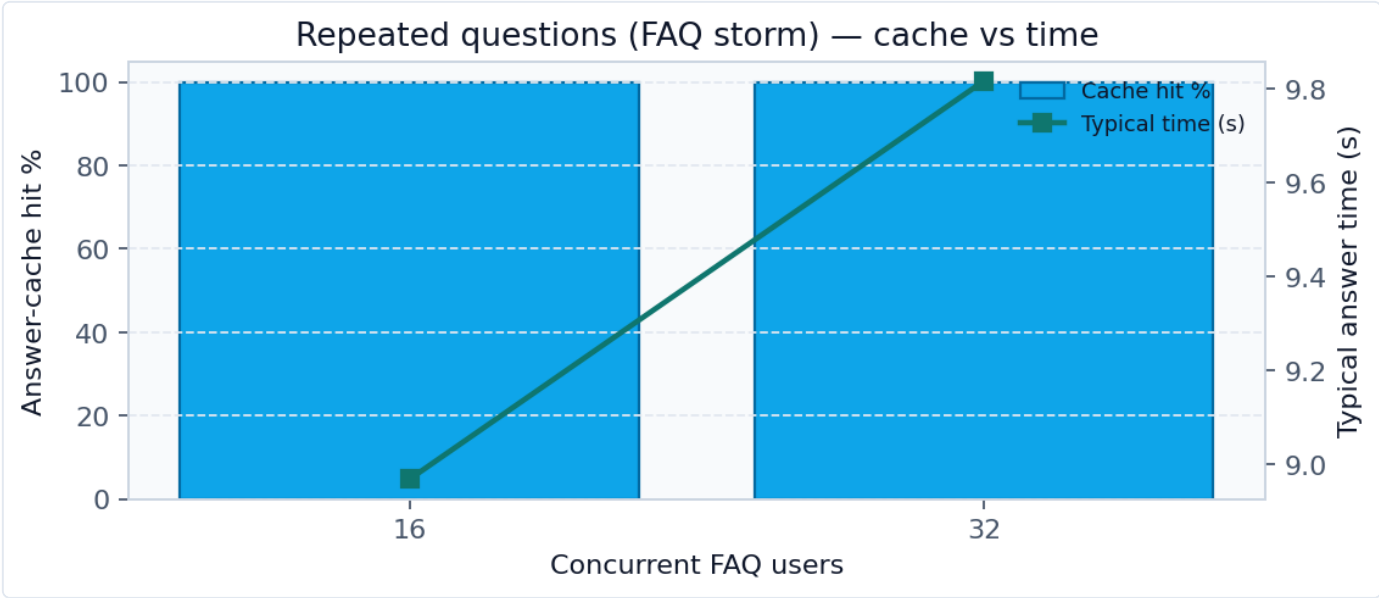


λ	OFFERED	DONE	IN SLO	TYPICAL	UPPER	ERRORS
0.25/s	15	15	93%	15.5 s	20.2 s	0%
0.5/s	30	30	40%	30.1 s	37.9 s	0%

Knee (H2): Last comfortable level: 0.25 asks/s at 93% attainment.

4. Repeated FAQ questions (H3)

Same three prompts asked many times — expects high answer-cache hit rate and much faster responses. Reported separately from cold capacity (LAW-153-6).



USERS	ASKS	IN SLO	TYPICAL	UPPER	CACHE HIT
16	16	100%	9.0 s	14.7 s	100%
32	16	100%	9.8 s	22.0 s	100%

5. What this does and does not mean

ALLOWED

“About N concurrent fresh askers stayed inside ~25s on this demo tenant/ workspace.”

ALLOWED

“Repeated FAQs mostly hit the answer cache and returned faster.”

FORBIDDEN

“The demo is rated for N users globally” without tenant/workspace/date.

- Not an Acc (answer-quality) score — see SPEC-001.
- Not a database micro-benchmark — see SPEC-090.
- No document upload or Vision OCR was run on the shared demo.
- Client machine details are inventory only — they are not SUT capacity.

6. Protocol & machines appendix

Full protocol: specs/153-workload-benchmark/07-highload-protocol.md. Machine inventory file: machines.json in this run directory.

FIELD	CLIENT	SUT
Role	load_generator	sut
Identity	Raphael-Mac-Book-Pro-Max.local	https://demo.edgequake.com
OS / build	27.0	product 0.28.3
Compute	16 CPU / 128.0 GB	GCP-hosted shared demo (deploy/gcp). Exact VM SKU may be opaque from outside; record only what /health and public deploy docs expose.
Auth	—	X-API-Key (secret not stored)
	—	['mistral-small-latest']

FIELD	CLIENT	SUT
Workspace		
LLM		

Generated by `bench153.render_highload_pdf` · charts via matplotlib (Agg) embedded as PNG.